Original Article

Development and validation of a diagnostic model for thoracic aortic dissection based on radiomics derived from non-enhanced computed tomography imaging

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Background: Thoracic aortic dissection (TAD) is a life-threatening disease, but the current conventional diagnostic methods are limited. In this study, we developed and validated a nomogram based on clinical features and non-enhanced computed tomography (CT) radiomics for the diagnosis of TAD.

Methods: A retrospective analysis of the data of 265 patients who underwent both non-enhanced and enhanced chest CT scans at two medical centers was conducted. The patients were randomly divided into a training set (n=147) and a validation set (n=63) at a 7:3 ratio. Patients from another institution were designated as the external test set (n=55). A multivariate logistic regression analysis was conducted to identify the independent clinical predictors and CT features. The least absolute shrinkage and selection operator (LASSO) logistic regression algorithm was used to select the optimal radiomic features.

Results: Combining the clinical independent predictors with the radiomics score (Rad score) enhanced the predictive power of the nomogram, which showed good consistency between the training and validation sets. In the training set, the nomogram had an area under the curve (AUC) of 0.968 [95% confidence interval (CI): 0.942–0.993], and an accuracy, specificity, and sensitivity of 92.5%, 95.9%, and 89.2%, respectively; while in the validation set, it had an AUC of 0.965 (95% CI: 0.919–1.000), and an accuracy, specificity, and sensitivity of 92.1%, 91.7%, and 92.6%, respectively.

Conclusions: The nomogram based on radiomic features and clinical characteristics from non-enhanced CT imaging can effectively diagnose TAD and could serve as a tool for screening high-risk patients with TAD.

Keywords: Aortic dissection (AD); multi-slice computed tomography (multi-slice CT); radiomics; receiver operating characteristic curve (ROC curve); nomogram

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Introduction

Aortic dissection (AD) is the most severe complication of thoracic aortic disease (1). The incidence of AD varies widely, ranging from approximately 3 to 16 cases per 100,000 people annually, depending on the study design and geographic region (2,3). If the condition is not managed within the first 48 hours of symptom onset, the mortality rate increases by 1% per hour (4-6). Thus, timely and accurate diagnosis is crucial for improving patient survival and preventing serious complications.

The true incidence of AD is higher than that currently reported, and as many as one-third of AD cases may go undiagnosed. In previous studies, approximately 5–15% of AD patients did not exhibit typical pain symptoms, which may lead to the stealthy progression and deterioration of the condition (7-9). Compared to contrast-enhanced examinations, which require the assessment of the patient's medical history, allergy history to contrast agents, and overall health status, non-enhanced computed tomography (CT) is more readily accessible, has a lower radiation dose, and fewer contraindications; thus, it could serve as a tool for screening AD (10). However, non-enhanced CT is less sensitive in distinguishing between true and false lumens associated with AD, and has a higher false-negative rate (11).

Radiomics is defined as the high-throughput extraction of numerous image features from medical images, which are then used to construct diagnostic, predictive, and prognostic models (12). This approach is most commonly applied in oncology, and to date has rarely been used in vascular diseases (13,14). To assist in the accurate clinical diagnosis and early identification of thoracic aortic dissection (TAD), and to enhance the diagnostic efficacy of non-enhanced CT, this study developed a nomogram based on radiomics features from non-enhanced CT and clinical characteristics, and evaluated its effectiveness in the diagnosis of TAD. We present this article in accordance with the TRIPOD+AI reporting checklist (available at https://qims.amegroups.com/article/view/10.21037/qims-2025-229/rc).

Methods

Study participants

The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. The study was approved by the Institutional Review Board of The Fourth Affiliated Hospital of Harbin Medical University {No. 2024[51]}, and the requirement of individual consent for this retrospective analysis was waived. Harbin Medical University Cancer Hospital was also informed and approved the study.

A retrospective analysis was conducted of the data of 1,213 patients who underwent enhanced CT scans at The Fourth Affiliated Hospital of Harbin Medical University from January 2020 to October 2023, and the Harbin Medical University Cancer Hospital from June 2016 to June 2024 (*Figure 1*). Patients were included in the study if they met the following inclusion criteria: (I) underwent an enhanced CT; (II) underwent a non-enhanced chest CT scan; and (III) had a time interval between non-enhanced CT and enhanced CT of less than 3 days (1). Patients were excluded from the study if they met any of the following exclusion criteria: (I) had undergone aortic surgery; (II) had CT images of insufficient quality for analysis; and/or (III) had incomplete clinical data. Patients diagnosed with TAD by enhanced CT were included in the case group, while an equal number of non-TAD patients, matched as closely as possible for baseline characteristics, were selected as the control group. Patients were also excluded from the control group if they had intramural hematomas (IMHs), mural thrombi, or penetrating ulcers, which are difficult to differentiate from AD.

In the assessment of non-enhanced CT, the evaluation was jointly conducted by two radiologists with more than 7 years of diagnostic experience each. If neither radiologist identified TAD in the non-enhanced CT scans, the case was classified as having an undetectable TAD on the non-enhanced CT imaging. The flow chart illustrating our study process is shown in *Figure 2*.

Image acquisition

All the CT scans were conducted using multi-slice spiral CT scanners from Toshiba (Aquilion ONE TSX-305A or 320-row, Tokyo, Japan), Siemens Force (Forchheim, Germany), or United Imaging 640 (Shanghai, China). The longitudinal coverage of aortic computed tomography angiography (CTA) ranged from the top of the aortic arch to below the iliac artery bifurcation. Typically, 60–90 mL of contrast media (iohexol injection containing 350 mg/mL iodine, GE Healthcare, Beijing, China) was injected into the anterior cubital vein at 4.5 mL/s, followed by 30 mL saline. Bolus tracking synchronized the injection with CT scanning. For non-enhanced CT, the average scan range extended from the thoracic inlet to the superior mesenteric artery with a slice thickness of 1–5 mm. The other parameters were as follows:

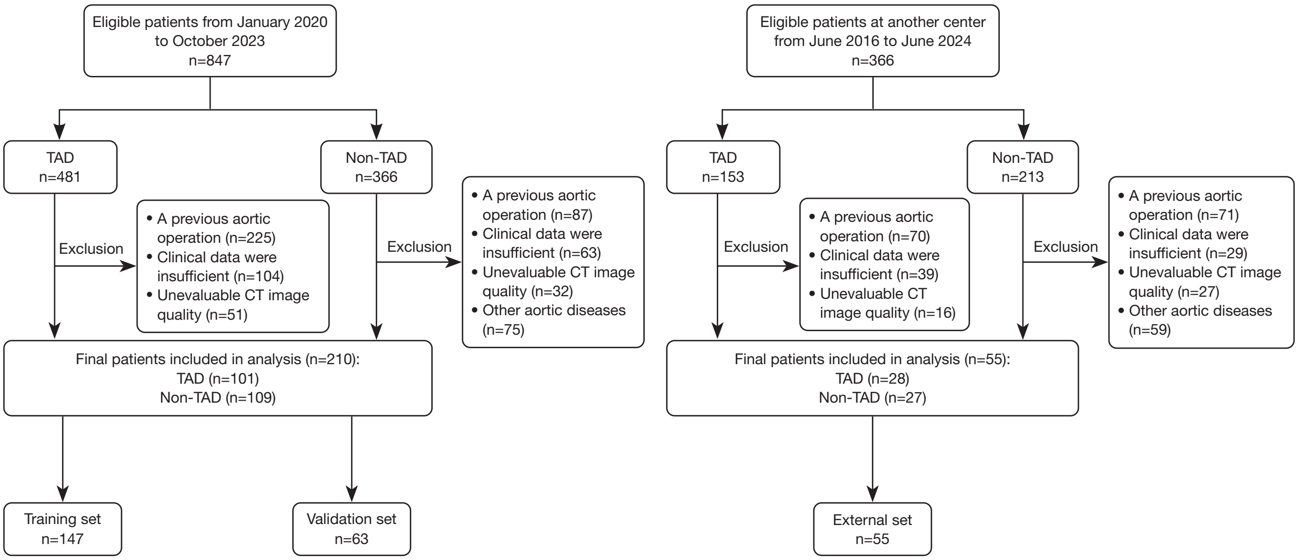

Figure 1 Flowchart showing recruitment pathway for patients. CT, computed tomography; TAD, thoracic aortic dissection.

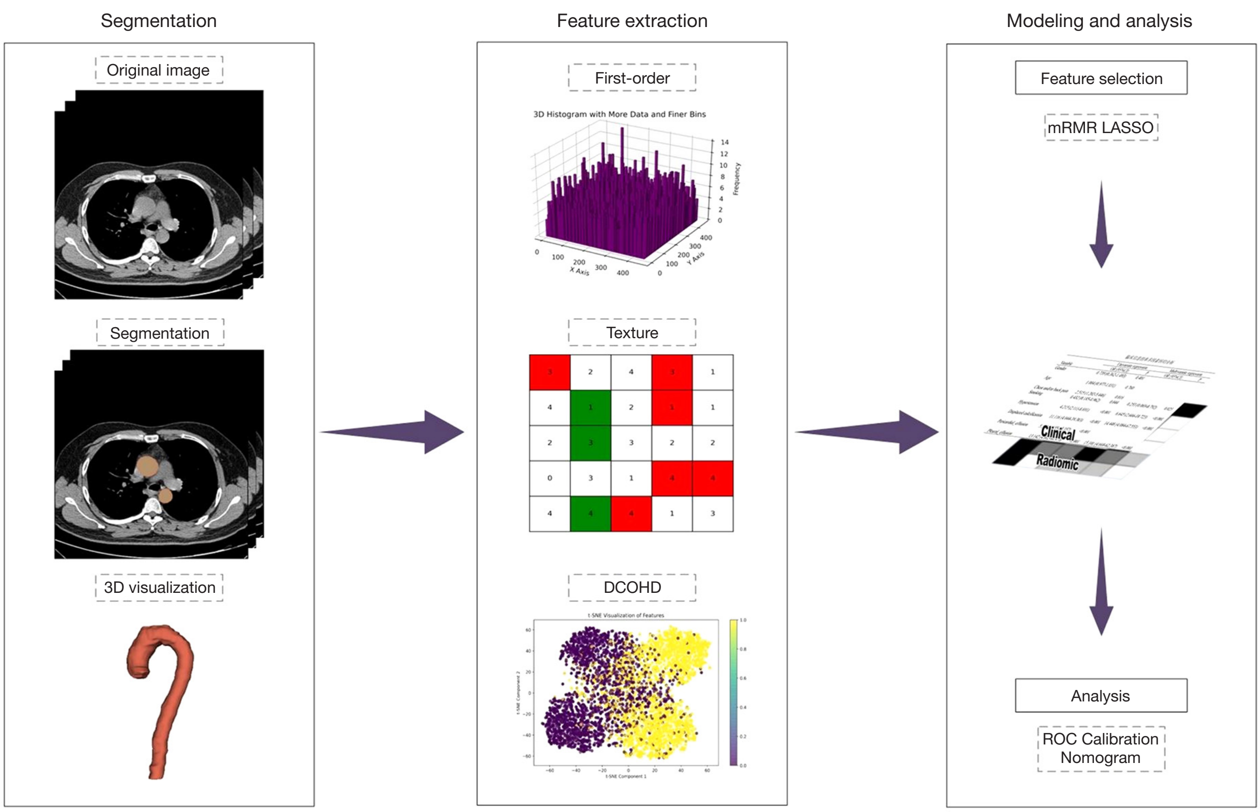

Figure 2 Workflow of radiomic analysis steps. 3D, three-dimensional; DCOHD, different co-occurrence of high-order descriptors; LASSO, least absolute shrinkage and selection operator; mRMR, maximum-relevance and minimum-redundancy; ROC, receiver operating characteristic.

a tube voltage of 100–120 kVp, tube current with automatic tube current modulation, and a matrix size of 512×512. For further details on the parameters, including collimation, pitch, and field of view (see Table S1). Images of the external test set were acquired using two additional multi-slice CT scanners from Philips (Amsterdam, Netherlands) and GE Healthcare, with a tube voltage of 120 kVp, automatic tube current modulation, and a slice thickness ranging from 1 to 5 mm. All the other imaging parameters were consistent with those of the Siemens Force CT scanner.

Segmentation and feature extraction

The volumes of interest (VOIs) in the non-enhanced CT images were manually segmented using three-dimensional Slicer software (version 5.6.2, http://www.slicer.org). This process was carried out by an experienced researcher, who first outlined the aorta on the cross-sectional images, and then automatically filled the aortic region extending from the aortic root to the diaphragm. The VOI range was subsequently manually adjusted to exclude irrelevant structures such as the myocardium, pleura, spine, peritoneum, and gastrointestinal tract as much as possible. Pyradiomics software (version 3.0) was used to extract a total of 1,409 features. The main radiomic features included the first-order and shape features, while the texture features included the gray-level co-occurrence matrix (GLCM), gray-level dependence matrix (GLDM), gray-level run length matrix (GLRLM), gray-level size zone matrix (GLSZM), and neighborhood gray tone difference matrix (NGTDM).

Feature selection and model construction

First, the Mann-Whitney test was applied to screen the features in the training set, and the results were adjusted using the Bonferroni method to control the error rate arising from multiple comparisons. Second, the maximum-relevance and minimum-redundancy (mRMR) method was employed to rank the features, and the top 20 features with the highest relevance were retained. Further, the least absolute shrinkage and selection operator (LASSO) logistic regression algorithm was used to reduce feature dimensionality, with the optimal regularization parameter λ selected through 10-fold cross-validation to determine the final feature subset. A linear combination of each feature and its coefficient was constructed for each patient to quantify the radiomics score (Rad score) of the radiomics; the Rad-score formula is detailed in Appendix 1. Univariable logistic regression was used to screen each clinical feature for statistical significance. Subsequently, the significant features from the univariable analysis underwent further refinement via multivariable logistic regression, ultimately yielding a set of clinically independent predictive factors. The Rad score and these clinically independent predictive factors were incorporated into a nomogram. The nomogram was constructed based on the training set, and validated using both the internal and external cohorts.

Statistical analysis

All the statistical analyses were conducted using R software (version 4.2.3). The *t*-test or non-parametric test were used to compare the continuous variables. The qualitative data are expressed as the number of cases (%), and groups were compared using the chi-square test to assess differences in clinical factors and CT characteristics between the TAD and non-TAD patients. All the statistical analyses were performed on a two-sided basis, with a P value of less than 0.05 considered statistically significant. The receiver operating characteristic (ROC) curve was plotted, and the area under the curve (AUC), accuracy, specificity, sensitivity, positive predictive value (PPV), and negative predictive value (NPV) were calculated to evaluate the predictive performance of the nomogram. The calibration curve was used to assess the calibration of the nomogram.

Results

Clinical characteristics

A total of 265 patients (median age: 59 years; interquartile range: 50–67 years; 177 males, 88 females) participated in the study. Of these, 129 were diagnosed with TAD (of whom, 70 had Stanford type A and 59 had Stanford type B), and 136 were diagnosed as non-TAD. Fifty cases of TAD were not detected on the non-contrast CT scans, resulting in a missed diagnosis rate of approximately 38.8%. The patients were randomly assigned to the training set (44 type A, 30 type B, 73 non-TAD; n=147), validation set (16 type A, 11 type B, 36 non-TAD; n=63), and external test set (10 type A, 18 type B, 27 non-TAD; n=55) using a random number table. The clinical factors and CT features of the patients in the training set, validation set, and external test set are shown in *Table 1*.

In the training set, significant differences between the TAD and non-TAD patients were observed for

Table 1 Clinical and imaging characteristics of the patients in the training, validation, and external test sets

Characteristics	Training set (n=147)			Validation set (n=63)			External test set (n=55)		
	TAD (n=74)	Non-TAD (n=73)	P	TAD (n=27)	Non-TAD (n=36)	P	TAD (n=28)	Non-TAD (n=27)	P
Age (years)	56.8±12.2	56.2±12.0	0.762	59.7±12.7	55.2±12.0	0.157	64.8±9.7	60.9±11.0	0.172
Gender			0.400			0.372			0.213
Female	25 (33.8)	20 (27.4)		9 (33.3)	16 (44.4)		7 (25.0)	11 (40.7)	
Male	49 (66.2)	53 (72.6)		18 (66.7)	20 (55.6)		21 (75.0)	16 (59.3)	
Chest and/or back pain			0.014			0.081			0.075
No	14 (18.9)	27 (37.0)		5 (18.5)	14 (38.9)		18 (64.3)	23 (85.2)	
Yes	60 (81.1)	46 (63.0)		22 (81.5)	22 (61.1)		10 (35.7)	4 (14.8)	
Smoking			0.041			0.797			0.631
No	63 (85.1)	52 (71.2)		21 (77.8)	27 (75.0)		18 (64.3)	19 (70.4)	
Yes	11 (14.9)	21 (28.8)		6 (22.2)	9 (25.0)		10 (35.7)	8 (29.6)	
Hypertension			<0.001			0.084			0.491
No	18 (24.3)	42 (57.5)		7 (25.9)	17 (47.2)		14 (50.0)	16 (59.3)	
Yes	56 (75.7)	31 (42.5)		20 (74.1)	19 (52.8)		14 (50.0)	11 (40.7)	
Displaced intimal flap			<0.001			<0.001			<0.001
No	54 (73.0)	73 (100.0)		14 (51.9)	36 (100.0)		12 (42.9)	27 (100.0)	
Yes	20 (27.0)	0 (0.0)		13 (48.1)	0 (0.00)		16 (57.1)	0 (0.0)	
Displaced calcification			<0.001			<0.001			<0.001
No	45 (60.8)	69 (94.5)		9 (33.3)	30 (83.3)		11 (39.3)	26 (96.3)	
Yes	29 (39.2)	4 (5.5)		18 (66.7)	6 (16.7)		17 (60.7)	1 (3.7)	
Pericardial effusion			<0.001			<0.001			<0.001
No	50 (67.6)	67 (91.8)		16 (59.3)	35 (97.2)		22 (78.6)	24 (88.9)	
Yes	24 (32.4)	6 (8.2)		11 (40.7)	1 (2.8)		6 (21.4)	3 (11.1)	
Pleural effusion			<0.001			0.005			0.022
No	42 (56.8)	69 (94.5)		17 (63.0)	33 (91.7)		19 (67.9)	25 (92.6)	
Yes	32 (43.2)	4 (5.5)		10 (37.0)	3 (8.3)		9 (32.1)	2 (7.4)	

Data are presented as mean ± SD or n (%). SD, standard deviation; TAD, thoracic aortic dissection.

hypertension, displaced intimal flap, displaced intimal calcification, pericardial effusion, and pleural effusion (P<0.001). In the validation set, there were significant differences in displaced intimal flap, displaced intimal calcification, pericardial effusion, and pleural effusion between the TAD and non-TAD patients (P<0.05). Similarly, in the external test set, significant differences were observed in displaced intimal flap, displaced intimal calcification, pericardial effusion, and pleural effusion between the TAD and non-TAD patients (P<0.05). The multivariable logistic regression analysis identified hypertension, displaced intimal calcification, and pleural effusion as independent predictors (P<0.05) for establishing a clinical model.

Radiomic feature selection

A total of 1,409 radiomic features were extracted, including 270 first-order features, 360 GLCM features, 210 GLDM features, 240 GLRLM features, 240 GLSZM features,

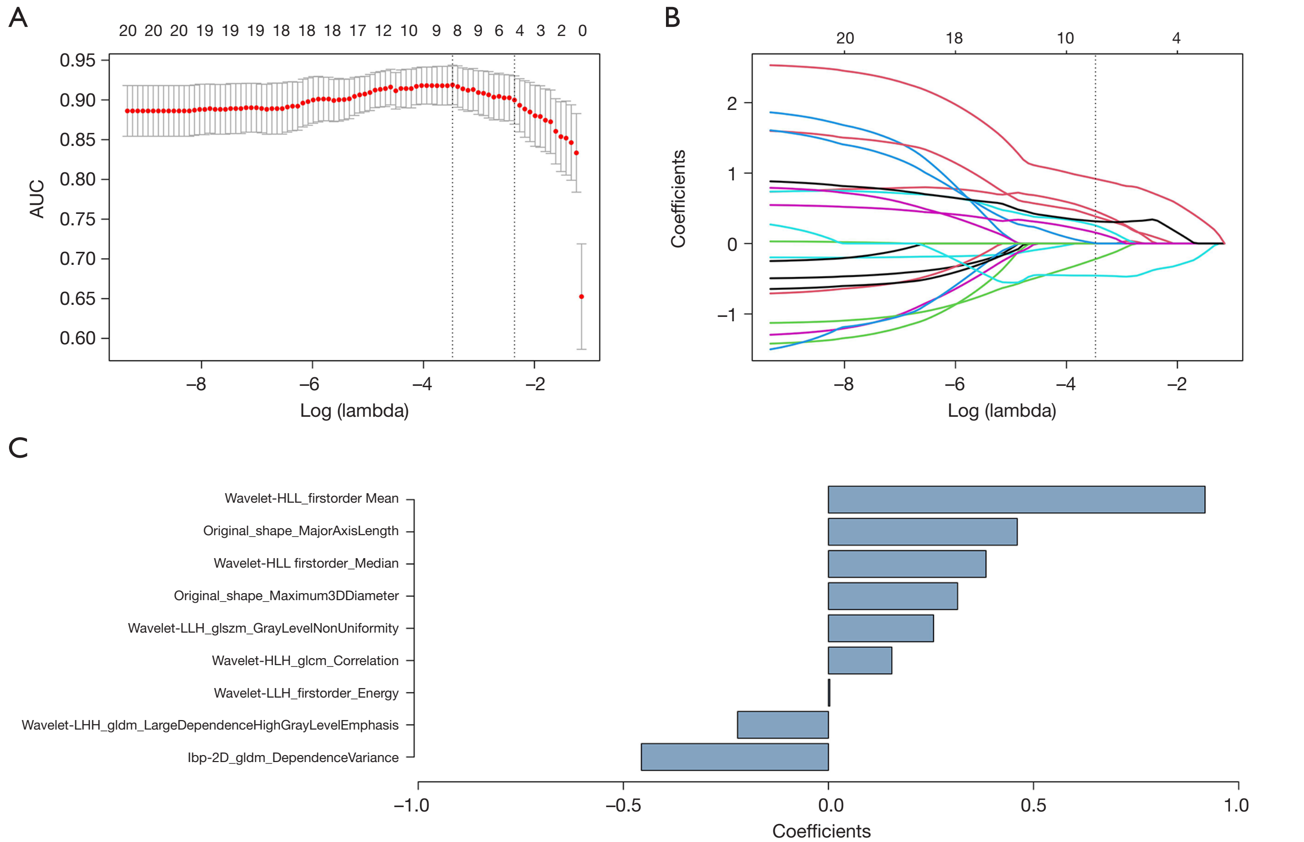

Figure 3 Non-enhanced CT radiomic feature selection. (A) Selection of the optimal tuning parameters by 10-fold cross-validation. (B) The regression coefficients of LASSO. Each colored line represents the variation curve of the characteristic coefficient with the lambda value. (C) The most predictive subset of features and the corresponding coefficients. CT, computed tomography; LASSO, least absolute shrinkage and selection operator.

75 NGTDM features, and 14 shape features. Using the mRMR method to eliminate redundant and irrelevant features, 20 features were retained. LASSO regression10-fold cross-validation was employed to determine the optimal values of the tuning parameter λ. Applying the LASSO logistic regression algorithm in the training set, nine features with non-zero coefficients were selected, including three first-order features, two shape features, two GLDM features, one GLCM feature, and one GLSZM feature. For details of the feature names and their corresponding weights, see *Figure 3*.

Radiomics model analysis

The radiomics model, based on logistic regression, was constructed from a linear combination of the nine selected features, each weighted by its coefficient. The model showed excellent predictive power for TAD across the three cohorts. In the training set, it achieved an AUC of 0.945 [95% confidence interval (CI): 0.911–0.979], and a sensitivity of 82.4%, specificity of 97.3%, accuracy of 89.8%, PPV of 96.8%, and NPV of 84.5%. In the validation set, it had an AUC 0.924 (95% CI: 0.861–0.977), and a sensitivity of 92.6%, specificity of 80.6%, accuracy of 85.7%, PPV of 78.1%, and NPV of 93.5%. In the external test set, it had an AUC of 0.854 (95% CI: 0.757–0.952), and sensitivity of 92.9%, specificity of 63.0%, accuracy of 78.2%, PPV of 72.2%, and NPV of 89.5% (*Table 2*).

Evaluation and verification of the nomogram

Based on the multivariable logistic regression analysis, hypertension, pleural effusion, and displaced intimal calcification were identified as independent risk factors for

Table 2 The diagnostic performance of the individual models

Model	AUC (95% CI)	Sensitivity (%)	Specificity (%)	Accuracy (%)	PPV (%)	NPV (%)
Radiomic model						
Training set	0.945 (0.911–0.979)	82.4	97.3	89.8	96.8	84.5
Validation set	0.924 (0.861–0.977)	92.6	80.6	85.7	78.1	93.5
External test set	0.854 (0.757–0.952)	92.9	63.0	78.2	72.2	89.5
Clinical model						
Training set	0.842 (0.783–0.901)	58.1	93.2	75.5	89.6	68.7
Validation set	0.830 (0.725–0.934)	66.7	88.9	79.4	81.8	78.0
External test set	0.847 (0.740–0.954)	82.1	88.9	85.5	88.5	82.8
Nomogram						
Training set	0.968 (0.942–0.993)	89.2	95.9	92.5	95.7	89.7
Validation set	0.965 (0.919–1.000)	92.6	91.7	92.1	89.3	94.3
External test set	0.909 (0.835–0.983)	67.9	96.3	81.8	95.0	74.3

AUC, area under the curve; CI, confidence interval; NPV, negative predictive value; PPV, positive predictive value.

TAD (P<0.001). The clinical model incorporating these factors achieved AUC values of 0.842 (95% CI: 0.783–0.901) in the training set, 0.830 (95% CI: 0.725–0.934) in the validation set, and 0.847 (95% CI: 0.740–0.954) in the external test set (*Table 2*). The nomogram, incorporating the Rad score and clinical risk factors, demonstrated superior performance compared to both the clinical model and the radiomics model alone in predicting TAD. *Figure 4* illustrates examples of the nomogram accurately distinguishing TAD in clinical practice. In the training set, the nomogram had an AUC of 0.968 (95% CI: 0.942–0.993), and a sensitivity of 89.2%, specificity of 95.9%, accuracy of 92.5%, PPV of 95.7%, and NPV of 89.7%. In the validation set, it had an AUC of 0.965 (95% CI: 0.919–1.000), and a sensitivity of 92.6%, specificity of 91.7%, accuracy of 92.1%, PPV of 89.3%, and NPV of 94.3% (*Table 2*). The missed diagnosis rate of the nomogram was calculated to be 10.8%. The ROC curves for the training set, validation set, and external test set are illustrated in *Figure 5*. The nomogram showed good performance in differentiating between type A TAD (AUC: 0.970) and type B TAD (AUC: 0.965) in the training set. See Figure S1 for the ROC curves.

Compared to the training and external test sets, the calibration curves for the combined model in the validation set did not align perfectly with the ideal scenario. This discrepancy might be due to the relatively small sample size of the validation set and potential measurement inconsistencies between different devices. Despite this, the Brier scores indicated high consistency between the predicted and actual diagnoses across all three datasets, reflecting the model's excellent predictive accuracy (*Figure 6*). DeLong's test revealed that in the training and external test sets, the AUC values of the nomogram were significantly different from the AUC values of both the clinical and radiomics models (train: P<0.001, P=0.016; external test: P=0.035, P=0.039). In the validation set, a significant difference in the AUC values was observed between the nomogram and the clinical model (P=0.006), while no such significant difference was observed between the radiomics model and the nomogram (P=0.061).

Discussion

The study enrolled a total of 129 TAD patients; TAD was not detected in 50 of these patients on non-enhanced CT scans, resulting in a missed diagnosis rate as high as 38.8%. By combining the clinical features with the Rad score, we constructed a nomogram diagnostic model that outperformed the use of clinical models or radiomics models alone, reducing the missed diagnosis rate to 10.8%. This finding shows the significant value of the nomogram as a potential tool for screening AD using non-enhanced CT.

CTA is the best imaging modality for the diagnosis of AD

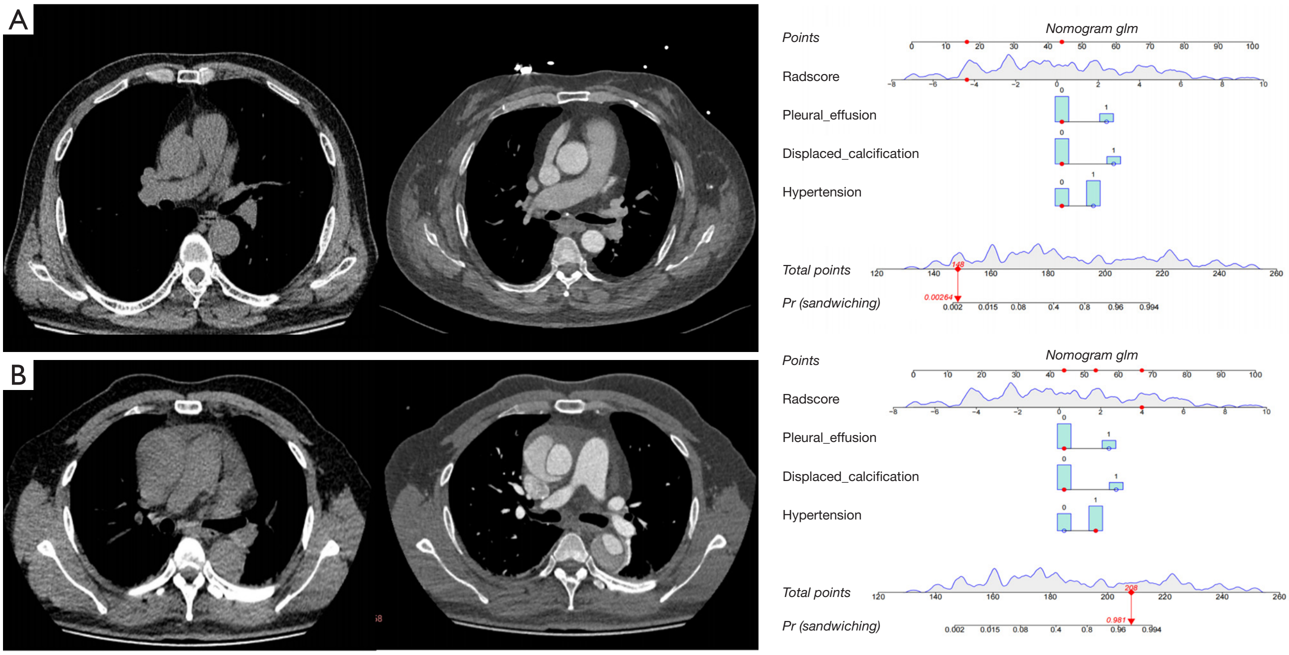

Figure 4 Examples of the use of nomograms in clinical practice. (A) A cross-sectional image from a non-enhanced CT and enhanced CT of a 59-year-old patient with a negative presentation. The nomogram indicates: pleural effusion =“no”, displaced calcification =“no”, hypertension =“no”, Rad score =–4.364, total score =148, with a probability of TAD of 0.00264. (B) A cross-sectional image from a non-enhanced CT and enhanced CT of a 50-year-old patient with Stanford type A aortic dissection. The nomogram indicates: pleural effusion = “no”, displaced calcification = “no”, hypertension = “yes”, Rad score =3.986, total score =208, with a probability of TAD of 0.981. CT, computed tomography; TAD, thoracic aortic dissection.

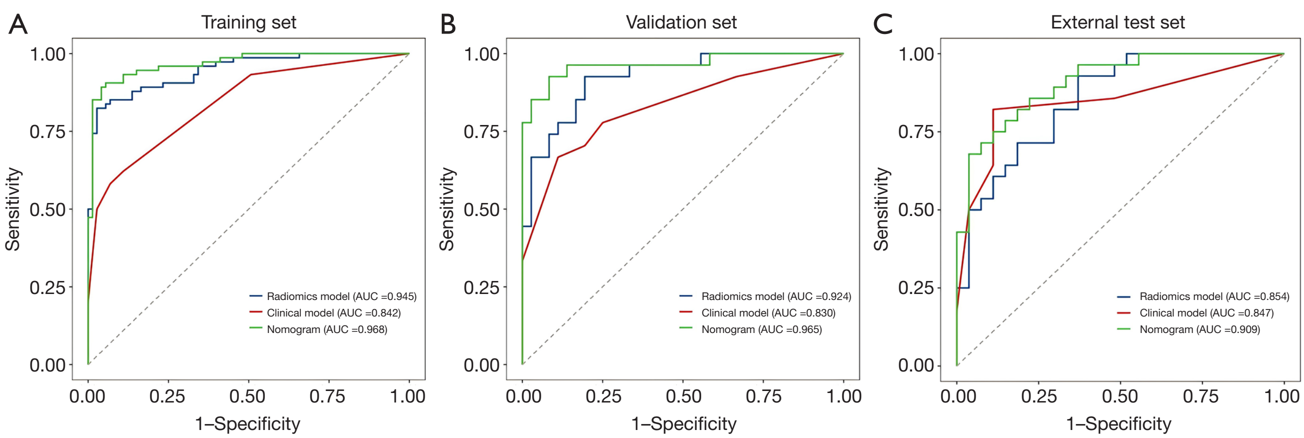

Figure 5 ROC curves of the radiomic model, clinical model, and nomogram. Comparison of ROC curves between the radiomic model (blue), clinical model (red), and nomogram (green) for the prediction of TAD in the (A) training, (B) validation, and (C) external test sets. AUC, area under the curve; ROC, receiver operating characteristic; TAD, thoracic aortic dissection.

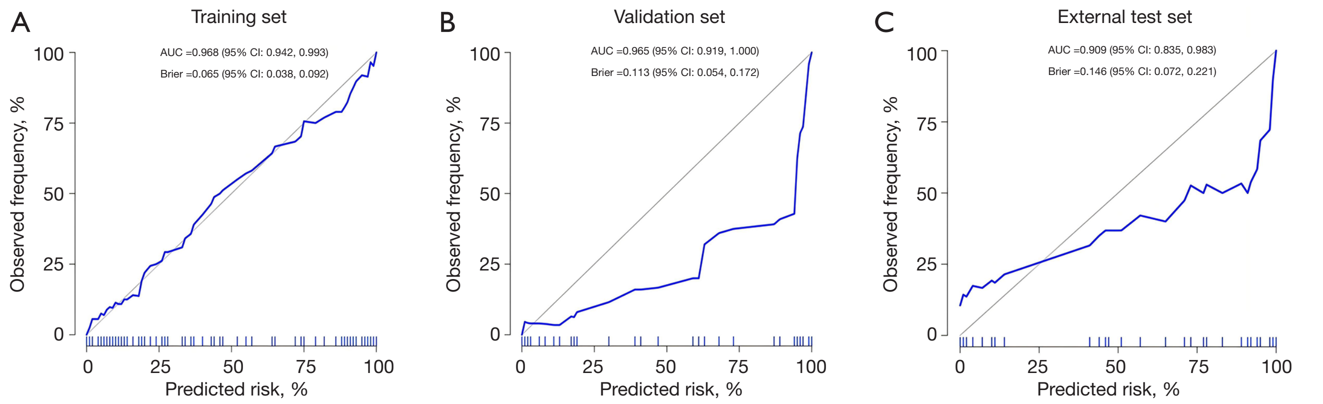

Figure 6 Calibration curves of the nomogram in each set (A-C). The 45° solid gray line represents a perfect prediction, and the other solid line represents the predictive performance of the nomogram in the training, validation, and external test sets. AUC, area under the curve; CI, confidence interval.

(15-17). However, given the urgent medical needs of AD patients, some healthcare organizations may find it difficult to provide round-the-clock emergency CTA examination services (18). In clinical practice, using non-enhanced CT for screening AD can help extend patient survival. However, the low sensitivity and high false-negative rate reported in previous studies need to be addressed. Notably, radiomics could aid in the diagnosis of TAD.

Radiomics has made significant progress in the field of tumor prediction; however, its application in the field of vascular disease remains limited. Cheng *et al.* developed a deep-learning algorithm based on non-enhanced CT radiomics, achieving automatic segmentation of aortic true or false lumens with 93.8% accuracy, demonstrating diagnostic performance comparable to radiologists (19). Hata *et al.* developed a deep-learning algorithm to detect AD using non-enhanced CT with 5 mm slice thickness, achieving a diagnostic level comparable to that of radiologists (20). Xiong *et al.* proposed a cascaded multitasking generation framework for the fully automated detection of AD using non-enhanced CT volumes. This framework outperformed existing models and is expected to significantly reduce the misdiagnosis rate of AD in clinical applications involving non-enhanced CT scans (21).

Additionally, several studies have focused on the diagnosis of acute aortic syndrome (AAS) through radiomics. Ding *et al.* predicted disease progression in aortic IMH patients using a radiomics model based on aortic CTA, which may improve clinical decision making (22). Guo *et al.* created a non-enhanced CT-based radiomics model to efficiently screen for TADs (23). However, their study had analytical limitations, as it did not incorporate clinical risk factors. The radiomics model developed by Zhou *et al.* showed accuracy in diagnosing acute AD. However, due to the limited sample size and lack of external validation in the study, the model's validity and generalizability require further evaluation and confirmation (24). We integrated data from two medical centers, collected the clinical information of 265 patients, and implemented an external validation process. The nomogram developed in our study achieved an AUC of 0.968 in the training set, which is higher than the AUC values reported in previous studies using radiomics on non-contrast CT for the diagnosis of AD.

In laboratory tests, an increase in D-dimer concentration is considered a potential indicator of AD, but its specificity is limited, as conditions such as pulmonary embolism and venous thrombosis can also elevate D-dimer levels (1). Asha *et al.* found that setting the D-dimer concentration threshold at 500 ng/mL achieved a sensitivity of 95.7% for diagnosing AD, but had a specificity of only 61.3%, which highlights the limitations of relying on a single diagnostic indicator (25).

Sudden and severe chest or back pain is a classic symptom of AD, and requires urgent attention. However, this article observed that among 129 patients with AD, as many as 46 (35.7%) lacked this typical pain characteristic, instead presenting with non-specific symptoms such as chest tightness, shortness of breath, palpitations, and syncope as their primary clinical manifestations. These patients are referred to as "asymptomatic" in this article. Among the 70 patients with type A AD, 19 were asymptomatic; similarly, of the 59 patients with type B AD, 27 were asymptomatic. After undergoing non-enhanced CT scans, these asymptomatic patients were screened using the nomogram developed in this study. The purpose of this screening approach is to promptly identify high-risk patients and initiate treatment measures to reduce mortality rates and the incidence of complications.

Limitations

This study had several limitations. First, there might be a potential selection bias in this study. All the patients enrolled in this study underwent both non-enhanced CT and enhanced CT, which might have resulted in a high rate of positive TAD. Second, although the non-enhanced CT data used in the study (derived from five different imaging devices) are close to clinical reality, the data inevitably introduce variability due to differences in imaging systems. Finally, the data for the external test set were obtained from a single oncology medical center, resulting in a relatively limited number of cases meeting the study criteria, which limits the breadth and depth of the conclusions. Therefore, a larger, multicenter study sample is needed in the future to further consolidate and expand the findings of this study.

Conclusions

The nomogram we constructed, which integrates Rad scores with clinical independent predictors, demonstrates superior diagnostic performance and could become a tool for screening high-risk patients with TAD, significantly optimizing the strategy for patient prognosis management.

Acknowledgments

None.

Footnote

Reporting Checklist: The authors have completed the TRIPOD+AI reporting checklist. Available at https://qims.amegroups.com/article/view/10.21037/qims-2025-229/rc

Data Sharing Statement: Available at https://qims.amegroups.com/article/view/10.21037/qims-2025-229/dss

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Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at https://qims.amegroups.com/article/view/10.21037/qims-2025-229/coif). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. The study was approved by the Institutional Review Board of The Fourth Affiliated Hospital of Harbin Medical University {No. 2024[51]}, and the requirement of individual consent for this retrospective analysis was waived. Harbin Medical University Cancer hospital was also informed and approved the study.

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Supplementary

Appendix 1

Radiomics score formula

Rad score = 0.0514540504064646 + original_shape_MajorAxisLength * 0.460319014852486 + wavelet-LHH_gldm_LargeDependenceHighGrayLevelEmphasis * –0.221495168474456 + wavelet-LLH_glszm_GrayLevelNonUniformity * 0.256174947442913 + wavelet-HLL_firstorder_Median * 0.384091593159151 + wavelet-LLH_firstorder_Energy * 0.00339104145031449 + lbp-2D_gldm_DependenceVariance * –0.455949011015411 + wavelet-HLH_glcm_Correlation * 0.154617591062485 + original_shape_Maximum3DDiameter * 0.314815967618912 + wavelet-HLL_firstorder_Mean * 0.918015527064413

Table S1 CT equipment parameters

Device name	Matrix	Collimation	Pitch	Slice thickness (mm)	Field of view (mm²)	Tube voltage (KV)	Tube current (mAs)
Aquilion ONE TSX-305A	512 x 512	0.5 x 80	0.813	1	256 x 256	120	Automatic Tube Medical University (ATCM)
Siemens Force	512 x 512	0.6 x 192	1.2	1	332 x 332	100	
Toshiba 320-row	512 x 512	0.5 x 100	0.87	1	400 x 400	120	
United Imaging 640	512 x 512	0.5 x 80	1.09	1	500 x 500	120	

All scanners were operated with the following settings: a tube voltage between 100–120 kV, and automatic tube medical university (ATCM).

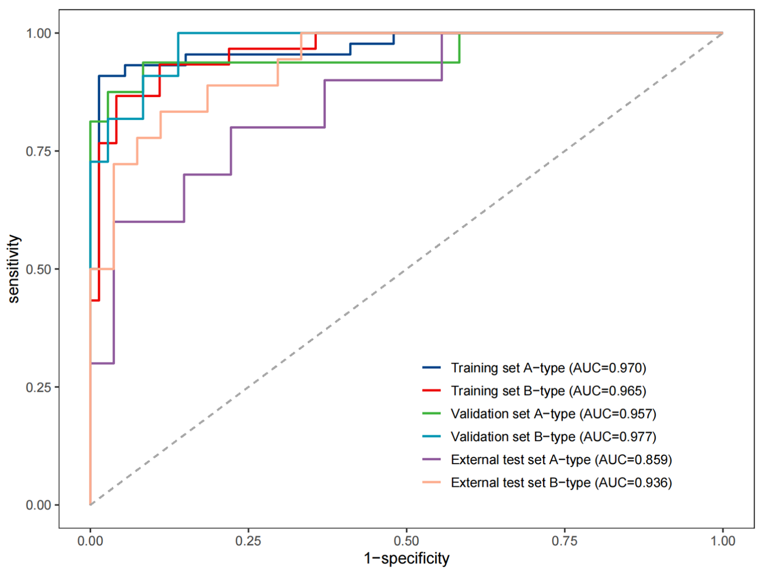

Figure S1 Assessment of the receiver operating characteristic (ROC) curves and area under the curve (AUC) as measures of the nomogram's diagnostic accuracy for Type A/B dissections in distinct cohorts.

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